

A Review of Silicon-Based Integrated Optical Switches

Xiaojun Chen, Jiao Lin, and Ke Wang*

Recently, silicon-integrated optical circuits have attracted intensive interests, thanks to the compatibility with the complementary metal-oxide-semiconductor (CMOS) technology that enables mass production at low cost. The optical switch is an essential part of optical integrated circuits, with broad applications in optical communications and networks, optical computing, and sensing such as LiDAR. In general, the silicon-integrated optical switch adopts thermo-optic or carrier dispersion effect to realize reconfigurable signal routing. However, the use of thermo-optic effect leads to high power consumption, and the carrier dispersion effect has the disadvantage of small refractive index change. In addition, both effects are non-latching, and hence, continuous power consumption is required even when switching is not needed. For overcoming these drawbacks, phase-change materials (PCMs) have been introduced into silicon-integrated optical switches. In this paper, silicon-integrated optical switches are classified according to the underlying structure and recent research is reviewed. Recent studies on silicon-integrated optical switches incorporating PCMs are also reviewed. Furthermore, the pros and cons of different types of integrated optical switches with and without PCMs are compared and discussed.

studied in silicon-based photonic integrated circuits,^[7] providing the basic building blocks for advanced functions.

The optical switch is a crucial part of optical integrated circuits,^[8–14] which are used to implement switching, routing, and optical cross-connections,^[15] with broad applications in optical communications and networks,^[16] optical computing,^[17] and sensing such as LiDAR.^[18] Different types of silicon-integrated optical switches have been investigated, which from the structure perspective mainly include the microring resonator (MRR) type,^[19–21] the Mach-Zehnder interferometer (MZI) type^[19,20,22,23] and the hybrid type which combines MRR and MZI.^[24] Switching speed,^[25] bandwidth,^[26] power consumption,^[27] extinction ratio,^[28] insertion loss,^[29] crosstalk,^[30] and footprint^[31] are important parameters of silicon integrated optical switches, and different applications have various requirements on these design parameters.

For example, for optical interconnection applications, wavelength selectivity and broad bandwidth are typically needed to support high-speed data transmission and to provide routing flexibility,^[32] while for optical computing, due to much more frequent switching operations, the power consumption and the switching speed are critical.^[33] Therefore, a wide variety of silicon-integrated optical switches have been studied and optimized in the literature in the past a few years.

The integrated optical switch based on silicon typically adopts the thermo-optic or the carrier dispersion effect to change the material refractive index for reconfigurable signal routing.^[34,35] However, the use of thermo-optic effect leads to high power consumption and relatively low switching speed, while the carrier dispersion effect is a weak effect with a small refractive index change, which leads to a large device length. In addition, both effects are non-latching (non-volatile) and hence, continuous power consumption is required during the steady state without switching operations, limiting the power efficiency.^[36–38] For overcoming these drawbacks, phase-change materials (PCMs) have been introduced into the silicon platform to realize integrated optical switches. PCMs are a unique kind of materials that show the latching property and a large change in optical properties in response to an external stimulus such as temperature, applied voltage, or ultrafast optical excitation.^[39,40] Taking advantage of these properties, PCMs have been incorporated with silicon-integrated circuits. By adjusting the PCM, both the real and imaginary parts (e.g., absorb light or allow light to pass) of

1. Introduction

The silicon photonic technology has been widely studied to realize high-density photonic integrated circuits using fabrication processes that are compatible with complementary metal-oxide-semiconductor (CMOS), which enables the mass production of photonic integrated circuits at low cost by leveraging existing CMOS electronic fabs.^[1–5] It has now become one of the most promising technologies in the photonic integrated circuit industry.^[6] Fiber-to-chip light couplers, power dividers, polarization splitters, waveguide crossings, optical filters, optical modulators, photodetectors and optical switches have all been widely

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its refractive index can be altered, realizing optical switches with reduced power consumption and footprint.

Currently, vanadium dioxide (VO_2)^[41] and $\text{Ge}_2\text{Sb}_2\text{Te}_5$ (GST)^[42] are the most common PCMs used in silicon-integrated optical switches. VO_2 is a phase change metal oxide that can be thermally tuned. At around 340 K, VO_2 undergoes a reversible transition from the insulator phase (low temperature) to the metal phase (high temperature),^[43] accompanied by reversible changes in optical, electrical, and magnetic properties.^[44,45] This transition can also be triggered by the optical way^[46] or the electrical way.^[47] In terms of the crystal structure, VO_2 changes from the monoclinic state VO_2 (M) at low temperature to the rutile state VO_2 (R) at high temperature.^[48] On the other hand, for GST, when its state changes between the amorphous state and the crystalline state, its optical and electrical properties undergo significant variation.^[49,50] This transition is fast,^[51] reversible,^[52] and can be triggered by stimuli such as short optical or electrical pulses.^[53–55] Its crystallization temperature for the phase transition is about 160 °C (433 K),^[56,57] and the amorphization temperature where the melting occurs is about 600 °C (873 K).^[39,57]

The silicon-integrated optical switch is highly attractive in a wide range of applications, and hence, it is a research field that has attracted intensive interest in the past a number of years, and various types of devices have been reported with different performances and focuses. Therefore, here we provide a review on this trending topic. Some review papers on silicon-integrated optical switches have been published recently. Chai et al. reviewed all-optical switching with a focus on different materials and the switching speed performance.^[58] However, other types of optical switches, such as electrical signal-driven switches, are not discussed. Focusing on the silicon photonic integration platform, Tu et al. provide a comprehensive review on the state-of-the-art silicon-integrated optical switches.^[20] However, it only focuses on pure silicon-based optical switches and does not discuss optical switches that further leverage PCMs on silicon. Miller et al.^[39] presents an in-depth review of PCMs in silicon-integrated photonic devices. However, the major focus is material properties of different types of PCMs. In addition, various types of devices using PCMs are reviewed, with only a small portion focusing on optical switches, limiting both the review breadth and depth. Different from previous review papers, in this paper, we discuss both pure silicon-integrated optical switches and silicon-integrated optical switches leveraging PCMs systematically, providing a comprehensive summary of this rapidly developing field.

This paper is organized as follows. We first classify the pure silicon integrated optical switches according to the underlying structure, and review related research works in Section II. Then, recent works on silicon-integrated optical switches further incorporating PCMs are reviewed in Section III. In Section IV, the advantages and disadvantages of various structures of pure silicon-integrated optical switches and silicon-integrated optical switches using PCMs are then compared, and possible ways to overcome the disadvantages are discussed. Section IV concludes this review paper.

2. Silicon-Integrated Optical Switches

In pure silicon-integrated optical switches, based on the device structure, the MRR type, the MZI type, and the hybrid type com-

bining MRR and MZI are the most common designs. Their phase shift mechanism can be classified into the thermo-optic (T-O) type, which changes the refractive index by varying the temperature via electrical heaters, and the electro-optic (E-O) type, which uses the carrier dispersion effect to change the refractive index via PIN junctions.^[20] In general, the optical switch using the T-O phase shift mechanism can achieve a smaller size, while the optical switch using the E-O phase shift mechanism can achieve lower power consumption and higher switching speed.^[38] In recent years, a large number of silicon-integrated optical switches have been proposed and studied. They are summarized in **Table 1** based on the underlying structure, and in the following subsections, we focus on reviewing these recent studies accordingly.

2.1. Silicon-Integrated Optical Switches Based on MRR

The MRR structure was first proposed by Marcanti in 1969, and it is composed of a ring-shaped waveguide coupled with a straight bus waveguide.^[80] Due to the advantages of high-quality factor (Q-factor), relatively small size, low power consumption, and wavelength selectivity, the MRR structure has been widely used in the design of silicon-integrated optical switches.^[74,81,82] In the past a few years, with the maturity of MRR devices, relatively large-scale silicon-integrated optical switches based on MRRs have become a research focus, and in this review, we focus on these recent studies.

In 2019, a scalable nonblocking 4×4 silicon integrated optical switch based on dual micro-rings was reported by Guo et al.^[71] For optical switches with multiple inputs and multiple outputs, they can be divided into the blocking and non-blocking types. The blocking type of switches^[74] cannot connect all permutation of sources and destinations, which leads to the decrease of data transmission efficiency and overall capacity. On the other hand, the nonblocking type supports establishing new connections between free inputs and outputs without disturbing existing connections.^[83] Hence, it is highly desirable for optical cross-connect to reduce the signal collision or loss probability,^[84] and it is adopted in this silicon-integrated optical switch study.^[71] Compared with the single MRR which has a sharp resonance peak, the dual MRR is more suitable for high-speed signals due to the flatter resonance peak.^[85] The basic structure of the switch is shown in **Figure 1a**, which has 6 groups of dual MRRs forming a 4×4 reconfigurable optical switch network that is connected by waveguides and crossings. In each group, the basic switch element consists of two micro-rings with the same resonance property, and it has two possible states of “bar” and “cross” states. **Figure 1b** shows the “bar” state, where the dual micro-rings do not resonate at the signal wavelength, and **Figure 1c** shows the “cross” state, where the signal wavelength is one of the resonant wavelengths of the dual microrings. The 4×4 optical switch has 2^6 states in total considering all basic switch elements. By configuring each basic switch element to a different state, light from one input port can be routed to any output port, realizing the complete switching capability. This switch was fabricated using the CMOS process on the SOI platform. The top silicon layer height was 220 nm and the buried silicon dioxide layer thickness was 2 μm . A 2.75 μm thick top SiO_2 layer was also deposited on top of the silicon devices using Plasma Enhanced Chemistry Vapor Deposition (PECVD).

Table 1. Comparison of various pure silicon-based integrated optical switches (MRR: micro-ring resonator; MZI: Mach-Zehnder interferometer; E-O: electro-optic; T-O: thermo-optic; EXP: experimental; SIM: simulation; BW: bandwidth; ER: extinction ratio; and IL: insertion loss).

Year	Structure type	Phase shift type	EXP/SIM	BW	Power	Speed	ER	Crosstalk	IL	Size	Comments
2006 ^[59]	MRR	E-O	EXP	–	–	–	About 15 dB	–	–	–	Limited extinction ratio
2012 ^[60]	MRR	E-O		About 100 GHz	37 mW	About 1 ns	About 50 dB	–	–	–	Limited bandwidth
2013 ^[61]	Hybrid	T-O		1.3 nm	–	About 7 μ s	–	–	–	–	Limited bandwidth, limited switching speed
2014 ^[62]	Hybrid	E-O		–	–	819 ps	–	<–20 dB	1.8 dB to 3.4 dB	–	E-O phase shift type leads to large device length
2015 ^[63]	Hybrid	T-O/E-O		35 GHz	T-O mode: 22.37 mW, E-O mode: 1.38 mW	–	–	–18.4 dB	–	–	E-O phase shift type leads to large device length
2015 ^[64]	MZI	T-O		–	–	–	50.4 dB	–	–	–	T-O phase shift type leads to high power consumption
2015 ^[65]	MRR	T-O		–	–	26.2 μ s	–	–	–	–	Limited switching speed
2016 ^[66]	MZI	T-O		–	104.8 mW	–	–	–7.2 dB	3.7 dB to 13.1 dB	–	High power consumption, large insertion loss
2016 ^[67]	MZI	T-O		140 nm	–	–	>20 dB	–	–	–	T-O phase shift type leads to high power consumption
2016 ^[68]	MZI	E-O		–	–	–	–	–34.5 dB	About 2 dB	–	Large crosstalk
2017 ^[69]	MZI	T-O		–	About 117 mW	21 μ s	–	>–18.8 dB	–	–	High power consumption, limited switching speed
2018 ^[70]	Hybrid	T-O/E-O		About 0.33 nm	0.34 mW	2.1 ns	–	–20.5 dB	About 10.6 dB $\pm$ 1.7 dB	–	Limited bandwidth
2019 ^[71]	MRR	T-O		>50 GHz	–	19.2 μ s	–	<2.5 dB	–	–	Limited switching speed
2019 ^[72]	MRR	T-O		–	–	–	–	–39 dB	–	–	Large crosstalk
2019 ^[73]	MZI	T-O/E-O		–	T-O mode: 34 mW π^{-1} , E-O mode: 7 mW π^{-1}	–	–	T-O mode: –29.1 dB, E-O mode: –19.4 dB	T-O mode: 1.74 dB $\pm$ 0.59 dB, E-O mode: 3.79 dB $\pm$ 1.32 dB	–	Large crosstalk
2020 ^[74]	MRR	T-O		>50 GHz	–	–	–	–	<5 dB	2.5 mm ²	T-O phase shift type leads to high power consumption
2020 ^[75]	MRR	T-O		165 GHz	–	–	–	–	–	4 mm ²	T-O phase shift type leads to high power consumption
2020 ^[76]	MZI	T-O		–	0.48 mW	530 μ s	–	–	–	–	Limited switching speed
2020 ^[77]	MZI	T-O		–	–	–	30 dB	–	–	–	T-O phase shift type leads to high power consumption
2020 ^[78]	MZI	T-O		4.2 nm	–	–	–	–	–	–	T-O phase shift type leads to high power consumption
2021 ^[79]	MZI	T-O		–	22.6 mW	0.4 μ s	–	–	–	60 $\times$ 30 μ m ²	T-O phase shift type leads to high power consumption

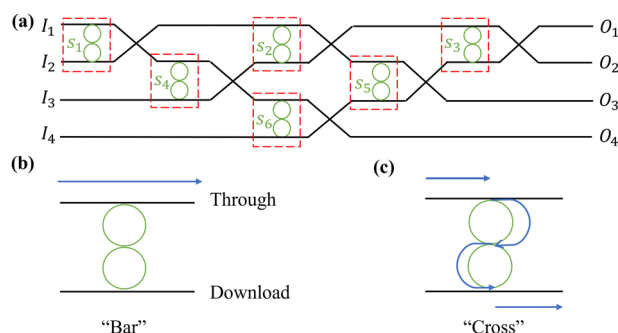


Figure 1. The structure of silicon integrated switch with dual micro-rings. a) Fabric of 4×4 reconfigurable optical switch, with each optical switch element formed from dual micro-ring resonators; b) the “Bar” state of each switch element; and c) the “Cross” state of each switch element. (re-drawn from^[71]).

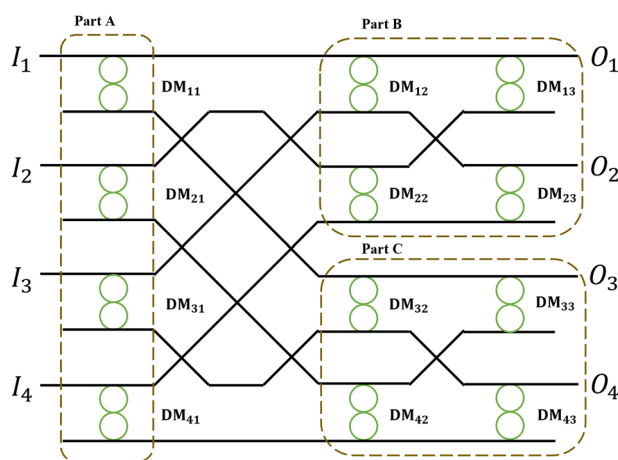


Figure 2. Fabric of the strictly nonblocking 4×4 optical switch array based on dual MRRs and the Butterfly topology. (re-drawn from^[74]).

The ridge waveguide width was 450 nm, the radius of MRRs was 10 μm , and the switch worked with the fundamental quasi-transverse electric (TE) mode. The insertion loss of this switch was better than 2.5 dB, together with an operation bandwidth better than 50 GHz. The rising and falling times are shown to be 4.9 μs and 14.3 μs (T-O effect), respectively. The key advantages of this switch include the scalability by adding more basic switch elements, the flat passband due to the use of dual microrings, and the relatively small number of basic switch elements required.

Using dual MRRs, Guo et al.^[74] further demonstrated a strictly non-blocking 4×4 optical switch array based on the Butterfly structure in 2020. The Butterfly structure can bring the benefits of using fewer switch elements and achieving balanced performance of loss and crosstalk. The optical switch array is shown in Figure 2, which can be decomposed to three parts. In the part A, the incoming optical signal is first switched to either part B or part C. For the signals switched to part B, four dual MRRs are used, and the signals can be further routed to the outputs O_1 or O_2 via the dual MRRs. Similarly, for the incoming signals originally switched to part C, they can be further routed to O_3 and O_4 ports. This switch was fabricated using similar processes and parameters as the prior study,^[71] with the same 450 nm width ridge waveguide and 10 μm radius microrings. The ring-to-ring

coupling gap in each basic switch element was selected to be 420 nm. 120 nm thick TiN layer was used as the microheaters to tune these dual micro-ring-based switch elements via the T-O effect. To deal with the thermal crosstalk caused by thermal conduction of silicon between micro-rings, heat insulating grooves were etched around each dual micro-ring. Measurement results show that the 4×4 switch array has the worst case on-chip insertion loss of smaller than 5 dB, a footprint of 2.5 mm^2 and a working bandwidth of over 50 GHz.

Different from using dual microrings, the dual add-drop micro-rings switching element was also proposed and a multi-stage 8×8 silicon photonic integrated switch was demonstrated in 2020 by Huang et al.^[75] As shown in Figure 3a, this switch arranges 12 switching elements into an Omega network to achieve connectivity between 8 inputs and 8 outputs. Figure 3b shows the basic 2×2 switching element based on the dual add-drop microrings, where each microring can be independently controlled to the “Bar” state or the “Cross” state. In addition to the independent control and operation, the dual add-drop micro-rings configuration can also be operated similarly to the dual microrings discussed above when both MRRs in a basic switch element are aligned in resonance. In this case, the resonance transmission passband can be widened. According to the results presented, the bandwidth was boosted from 120 GHz when a single MRR was used to 165 GHz when both MRRs were aligned in resonance. Hence, better routing flexibility can be achieved using the dual add-drop microrings configuration. This switch was fabricated and experimentally characterized. Results show the on-chip loss is between 4.4 and 9.6 dB end-to-end, and the worst-case average crosstalk leakage is -16 dB. The end-to-end passband is over 55 GHz. With efficient doped waveguide T-O phase shifters, the tuning efficiency is shown to be 48.85 GHz mW^{-1} , and the switch fabric shows a reconfiguration time at the rise and fall edges of 1.2 μs and 0.5 μs , respectively. The footprint is 4 mm^2 .

To provide complete connectivity between inputs and outputs whilst reducing the crosstalk, Cheng et al.^[72] proposed a scalable MRR based silicon Clos optical switch fabric with switch-and-select stages in 2019. Figure 4a shows the basic structure of the switch-and-select topology. This is an $N \times N$ device consisting of two linear arrays, which are $1 \times N$ and $N \times 1$ spatial demultiplexers (DeMux) and multiplexers (Mux), respectively. These two linear arrays are connected by a perfect shuffle network. Therefore, a total of $2N^2$ micro-ring switching cells are required. Because each pair of input and output switching DeMux/Mux has a dedicated signal path, this architecture provides strictly non-blocking connectivity. The scale-up of this design only requires adding bypass rings in linear switching arrays, and hence, the scaling overhead is small. In addition, the first-order crosstalk leakage from the input switching array, which is outlined by yellow arrows in the figure, is significantly reduced by the off-resonance microrings at the output stage, and thus, only second-order crosstalk occurs, as represented by red arrows. By contrast, Figure 4b shows the typical crossbar topology of silicon-integrated optical switches. Whilst this design requires a smaller number of switching cells (N^2), it suffers from the first-order crosstalk, which limits the signal transmission performance in interconnects. The low crosstalk advantage of the switch-and-select optical switch topology is experimentally verified. Measurement shows that the crosstalk ratio of a 4×4 device is in the range

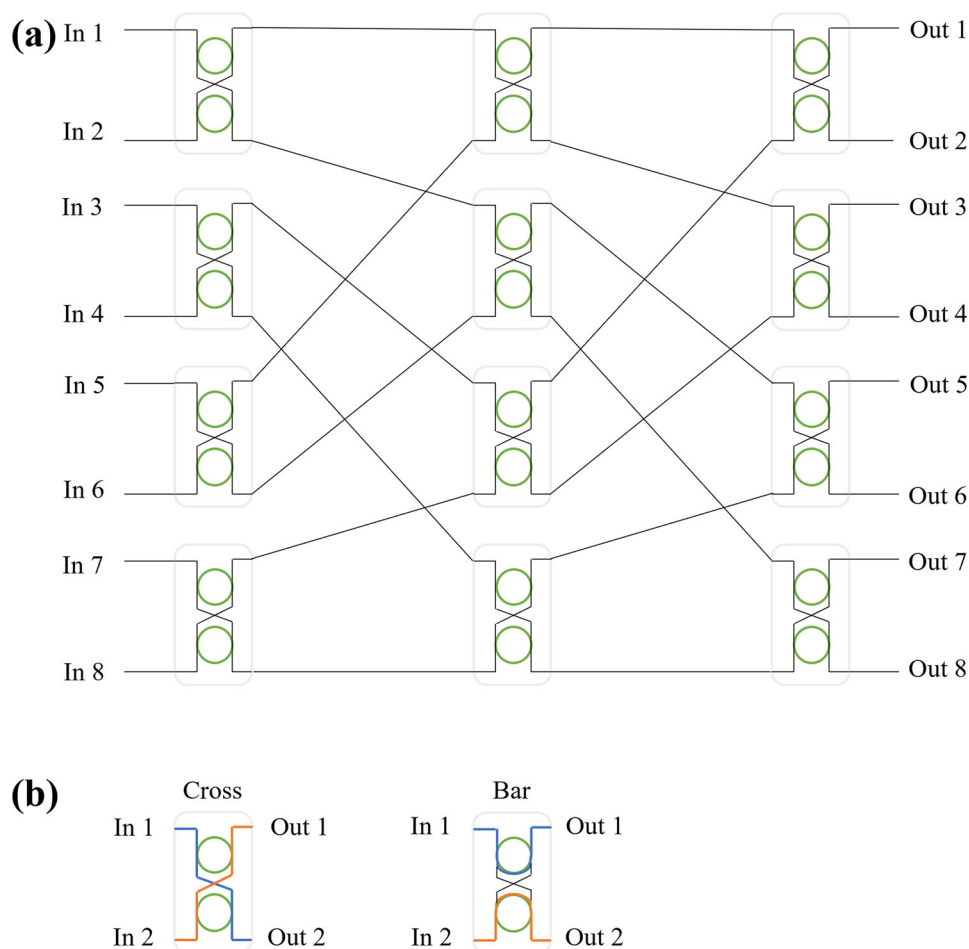


Figure 3. a) Structure of the 8×8 silicon integrated optical switch based on the Omega network; and (b) of the working states and configurations of basic switching element. (re-drawn from^[75]).

of -57 to -48.5 dB, enabling better than -39 dB crosstalk for the 16×16 switch by fully blocking the first-order crosstalk. In addition, compared with the single-stage switch-and-select fabric, the three-stage Clos design reduces the number of switch cells by up to 4 times.

2.2. Silicon-Integrated Optical Switches Based on MZI

In addition to MRRs, the MZI is also a commonly used structure in silicon-integrated optical switches. It is proposed in 1890s by Zehnder and Mach,^[86] which splits the input signal into two branches and then recombine them after passing through dedicated paths.^[87] By controlling the phase difference between two branches, constructive or destructive interference occurs upon recombination, realizing the signal switching function. Compared with the MRR structure, the MZI structure has a larger optical bandwidth and is less sensitive to temperature variations.^[88] Due to these advantages, MZIs have been used in silicon-based optical integrated circuits since the 1980s.^[89]

In 2019, a silicon-integrated strictly nonblocking 4×4 optical switch was demonstrated by Lu et al.^[73] The optical switch

is based on the double-layer network (DLN) architecture. It is composed of twelve 2×2 MZI switch elements and can be operated in both T-O and E-O switching modes. The structure of the 2×2 MZI switch element is shown in Figure 5a, which is based on two 2×2 symmetrical multimode interferometers (MMIs) and two active waveguide arms. The switch element supports broadband operation. Both TiN micro-heaters and p-i-n diodes are integrated in the waveguide arms to achieve both T-O and E-O switching in the same device. This means that the operation mode can be easily switched according to practical requirements: when the main requirements are small insertion loss and low power consumption, the T-O effect can be used; and when the main requirement is fast switching speed, the E-O effect can be activated. Next to the waveguide arms, two air trenches are etched, for suppressing thermal crosstalk and lowering the T-O power consumption. Figure 5b shows the schematic of the 4×4 DLN optical switch. This optical switch is fabricated based on the CMOS process, where the common SOI platform with 220 nm thick top silicon layer and 2 μm thick buried oxide layer is used. The footprint of the optical switch is $4.6 \text{ mm} \times 2 \text{ mm}$, including all the electrical pads and the input/output gratings. For each MZI element, the footprint is $70 \mu\text{m} \times 640 \mu\text{m}$, and the pitch

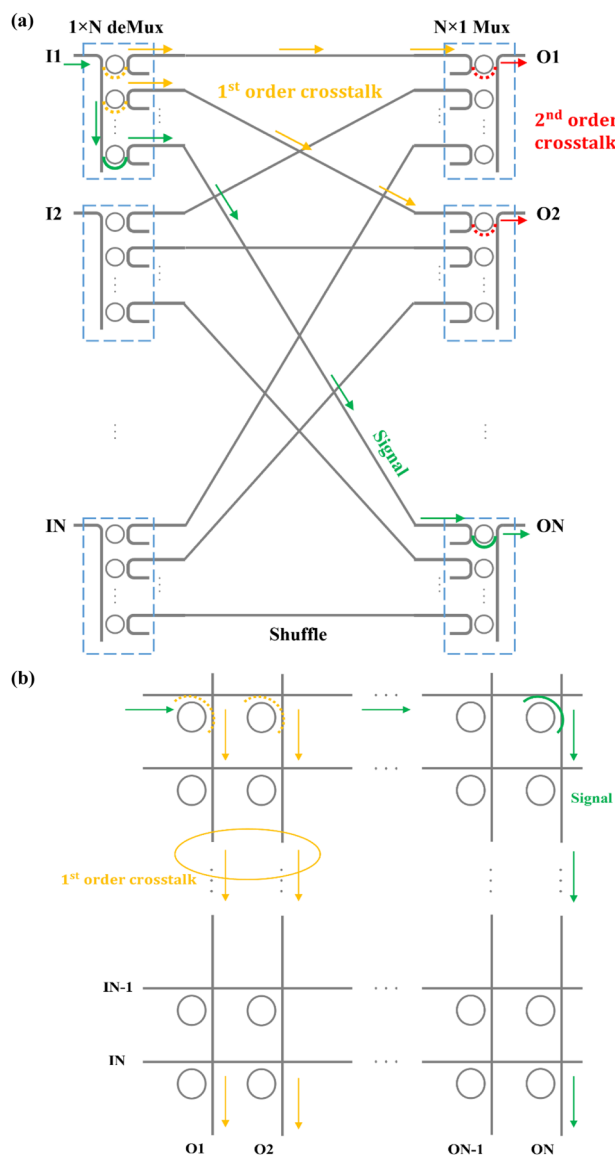


Figure 4. a) Schematic of the MRR-based switch-and-select topology, where the signal (green lines) routes from input 1 to output N , the yellow lines show the first order crosstalk, and the red lines show the second order crosstalk; and b) schematic of the micro-ring-based crossbar topology, where the signal (green lines) routes from input 1 to output N , and the yellow lines show the first order crosstalk. (re-drawn from^[72]).

sizes between MZIs are $120\ \mu\text{m}$ and $1040\ \mu\text{m}$ in the longitudinal and lateral directions, respectively. For the T-O switching mode, the power consumption is $34\ \text{mW}\ \pi^{-1}$, the on-chip insertion loss is $1.74 \pm 0.59\ \text{dB}$, the crosstalk is $-29.1\ \text{dB}$, and the temporal response rise time and fall time are $56\ \mu\text{s}$ and $16\ \mu\text{s}$, respectively. On the other hand, when the E-O switching mode is activated, the power consumption is $7\ \text{mW}\ \pi^{-1}$, the on-chip insertion loss is $3.79\ \text{dB} \pm 1.32\ \text{dB}$, the crosstalk is $-19.4\ \text{dB}$, the rise time is $3.2\ \text{ns}$, and the fall time is $2.5\ \text{ns}$.

To optimize both power consumption and device size, Wang et al.^[76] proposed a low power and compact T-O silicon integrated optical switch based on the MZI structure in 2020, where the Ti

heater was used to alter the operation temperature and introduce the phase difference between two arms. The switch is designed on a SOI wafer, which has a $220\ \text{nm}$ thick silicon layer on top of a $3\ \mu\text{m}$ thick buried oxide layer. As shown in Figure 6, in the MZI bias region, a 90° phase difference is introduced to ensure the MZI works in the linear region. The phase arm, based on $450\ \text{nm}$ wide waveguide, is isolated by removing the adjacent SiO_2 and underlying silicon to form an air trench. For the Ti heater, the width (W_h) is $1.5\ \mu\text{m}$, the thickness (t_h) is $100\ \text{nm}$ and the length (L_h) is $450\ \mu\text{m}$. For the air trench, the width (W_T) is $2\ \mu\text{m}$ with the same length of $450\ \mu\text{m}$. The gap between the heater and the air trench (G_{HT}) is $2\ \mu\text{m}$. For reducing the thermal conduction path from the heater to the phase arm, the thickness of SiO_2 cladding (t_c) is set to $500\ \text{nm}$. With these designs, a low switching power of $0.48\ \text{mW}$ is achieved at $1550\ \text{nm}$ for the TE mode. The 10%–90% response time of the switch is $530\ \mu\text{s}$, where the rise time is $150\ \mu\text{s}$ and the fall time is $380\ \mu\text{s}$.

To improve the switching speed, a silicon-integrated MZI-based optical switch with periodic electrodes and integrated heat sink was proposed in 2021 by Kita and Mendez-Astudillo^[79] The MZI optical switch loaded with two MMI phase shifters is shown in Figure 7, where the left arm MMI waveguide is lightly p-type doped ($1 \times 10^{18}\ \text{cm}^{-3}$), and the right arm MMI waveguide is undoped. In the middle of two MMI phase shifters, electrical connection is formed by the heavily doped Si, and the current flows through the two phase-shifters connected in series when a voltage is applied to the aluminum electrode, which is placed on top of the heavily doped Si region. The non-doped MMI heater consumes about 10 times the power of the doped MMI at the same current level. Therefore, this asymmetrically doped structure mainly heats the MMI waveguide on the right side, leading to a temperature difference between the two paths at left and right. In this way, phase difference is introduced to the two arms to switch the optical light intensity. This optical switch is fabricated using a CMOS foundry. The single-mode silicon waveguide has a dimension of $400\ \text{nm} \times 210\ \text{nm}$, and the heat sink is formed by TiN. Using this structure, which has a footprint of $60 \times 30\ \mu\text{m}^2$, a switching speed of $0.4\ \mu\text{s}$ and a power consumption of $22.6\ \text{mW}$ have been achieved.

The previously reviewed MZI-based silicon-integrated optical switches are wavelength independent. In 2020, Ikeda et al.^[77] proposed a novel silicon integrated wavelength selective switch based on sidewall-corrugated contra-directional couplers (C-DCs). The C-DC consists of two coupled waveguides with a periodic perturbation, and it has an ultra-broad free spectral range (FSR).^[78] The device design and working principle are shown in Figure 8. Figure 8a shows the basic structure of the C-DC, which is consisted of two adjacent waveguides with widths of W_1 and W_2 , respectively. The gap between the two waveguides is G , and the length is L . The two adjacent waveguides have sidewall corrugations with period Λ , and modulation depths of ΔW_1 and ΔW_2 , respectively. The blue arrows in Figure 8a show the contra-directional cross coupling between the propagating modes in each waveguide when the wavelength λ_B satisfies the Bragg condition of $\lambda_B = (n_1 + n_2)\Lambda$, where n_1 and n_2 are effective indices of the two waveguides, respectively. Based on C-DC, 2×2 single channel wavelength selective switch can be realized, and its structure is shown in Figure 8b, where C-DCs are used to drop and add signals as wavelength filtering devices, and the Mach-Zehnder

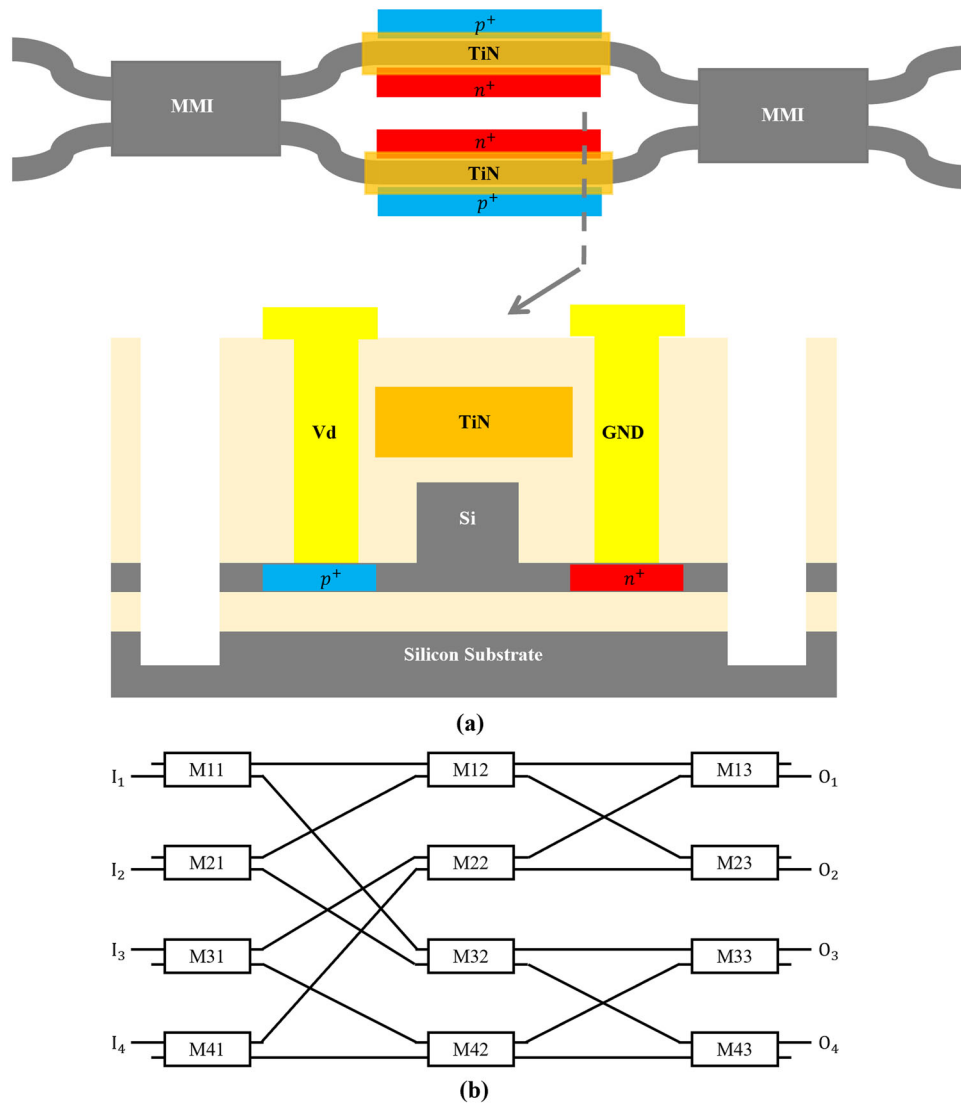


Figure 5. a) Structure of the 2 × 2 MZI switch element; and b) schematic of 4 × 4 DLN optical switch. (re-drawn from^[73]).

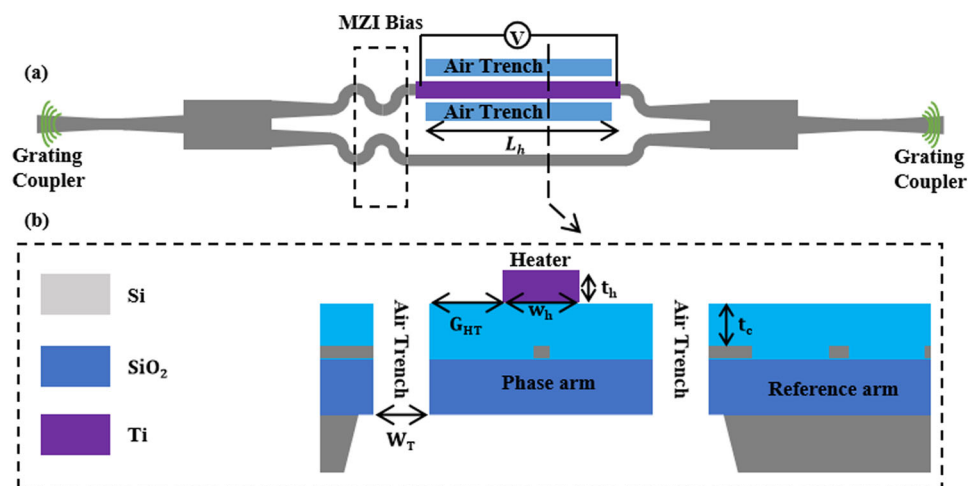


Figure 6. Schematic of the 1 × 1 thermo-optic switch. a) The top view; and b) the cross-section view along the dashed line in (a). (re-drawn from^[76]).

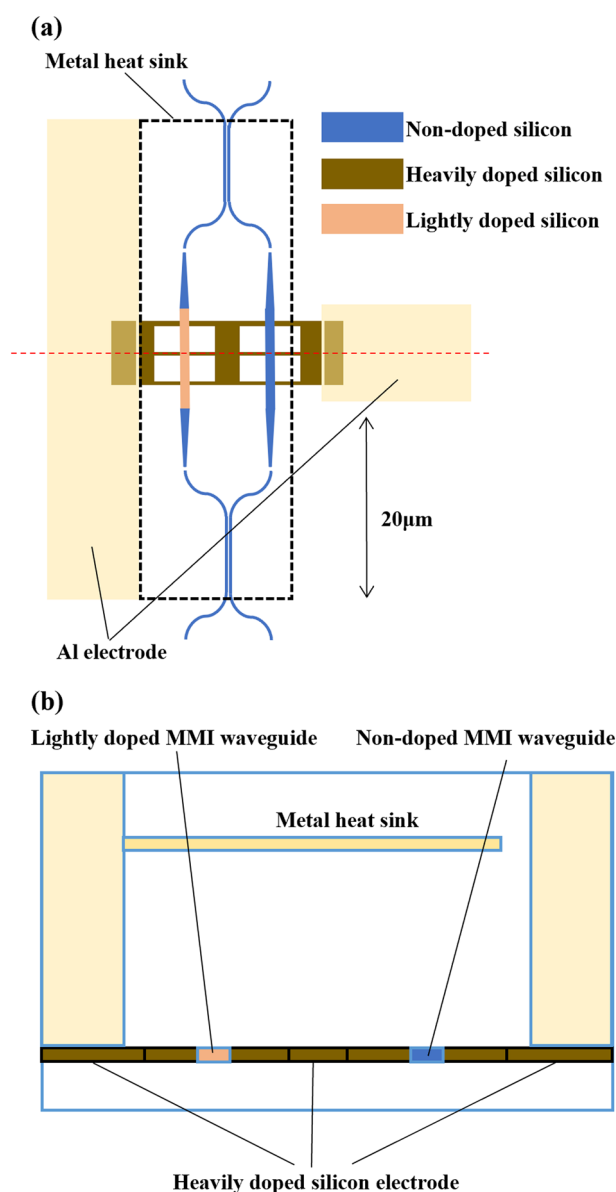


Figure 7. a) Schematic of the MZI optical switch with asymmetric phase shifters; and b) the cross-sectional view at the red dotted line in (a). (re-drawn from^[79]).

switch (MZ switch) is used to select the output port for the signal dropped by C-DCs. The MZ switch is shown in Figure 8d, where heaters are deposited on top of the two arms to control the phase difference. In this silicon-integrated wavelength selective switch, the wavelength channel with λ_B that enters the input port 1 is coupled to the other waveguide by the C-DC on top and then guided to the 2×2 MZ switch. When the MZ switch is in the cross state, the λ_B channel is guided to the lower C-DC from the right side, and hence, it is further cross-coupled by the lower C-DC to the output port 2'. When the MZ switch is in the bar state, the λ_B channel is guided by the MZ switch to the other side of the upper C-DC and then coupled back to the output port 1'. This working principle can be applied to all wavelengths satisfying the Bragg

condition. When signals with multiple wavelengths are injected to the input port, only channels satisfying the Bragg condition are back-coupled by the C-DCs and switched by the MZ switch, whilst the other channels pass through the silicon-integrated device directly.

The silicon-integrated switch based on C-DCs and MZ switches can also be extended to support more input and output ports, and one such device with N input ports is shown in Figure 8c, which is a $N \times N$ multi-channel (N_λ) wavelength selective switch. By increasing the number of C-DCs and the scale of the $N \times N$ MZ switch, the port count of the wavelength selective switch can be increased. By cascading single channel $N \times N$ wavelength selective switches with different Bragg wavelengths ($\lambda_1, \lambda_2, \dots$) in the horizontal direction, the number of wavelengths that can be selectively switched can be increased, providing good flexibility and scalability. This 2×2 switch was fabricated by a 300 mm CMOS R&D foundry using 45 nm ArF immersion lithography. The TE-like mode is adopted in the C-DCs design, and around 1550 nm wavelength is selected as the Bragg wavelength λ_B . For the nominal structure parameters, they are selected at 350 nm for W_1 , 450 nm for W_2 , 360 nm for Λ , 30 nm for ΔW_1 , 50 nm for ΔW_2 , 250 nm for G , and about 400 μm for L . The sidewall corrugation is also Gaussian-apodized. The fabricated 2×2 single-channel wavelength selective switch shows an out-band extinction ratio of at least 30 dB over a wide operation wavelength range of 70 nm from 1530 to 1600 nm.

In the previous work, only one wavelength is supported. Ikeda et al.^[78] further extended the number of wavelength channels to 16 over a broad wavelength range spanning across the C- and L-bands. The wide wavelength range is enabled by the ultra-broad FSR property of C-DCs. According to measurement results, this device has a fiber-to-fiber insertion loss of 9.2 dB, an on-chip loss of 5.4 dB, a 3-dB bandwidth of 4.2 nm for each wavelength channel (16 in total), a bar-port extinction ratio of 15.0 dB, and a cross-port extinction ratio of 23.0 dB. Furthermore, the average triggering power is 3.8 mW for the bar state and 15.6 mW for the cross state. The standard deviation of the triggering power amongst different wavelength channels (16 channels in total) is $\sigma = 2.1$ mW for the bar state and $\sigma = 2.3$ mW for the cross state. Hence, the capability of combining C-DCs and MZ switches to realize integrated ultra-broadband and multichannel wavelength selective switch or wavelength cross connect (WXC) switch is demonstrated.

2.3. Silicon Integrated Optical Switches Based on MRR and MZI Hybrid Structures

In the previous subsections, we have reviewed recent studies on silicon-integrated optical switches based on the MRR and MZI structures. For MRR-based switches, it has the superiorities of compact size and low operation power, whilst the passband, FSR, and operation stability (e.g., drifting of resonant wavelength) are limited. On the other hand, the MZI-based silicon integrated switches have a broad FSR and flat passband. However, the arms are typically long, and a high operation power is required. In order to further improve the performance, silicon-integrated optical switches combining MRR and MZI structures have been further studied to leverage the advantages of both types.

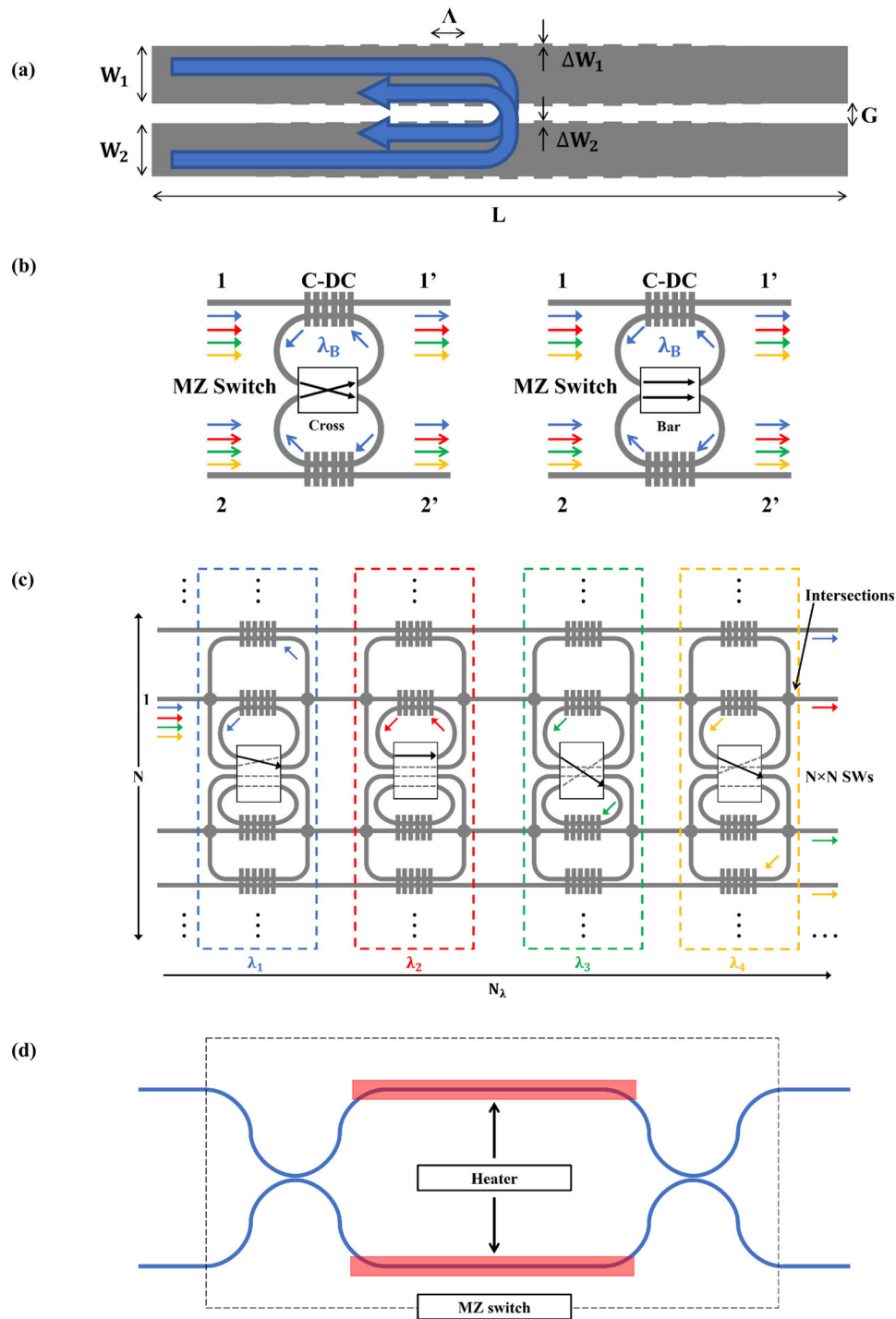


Figure 8. The schematic and working principles of silicon integrated optical switch with C-DCs. a) Structure of C-DC (the blue arrows indicate the cross coupling at λ_B); b) principle of 2×2 single-channel wavelength selective switch, where the left is the Cross state and the right is the Bar state; c) principle of $N \times N$ multi-channel (N_λ) wavelength selective switch, where blue, red, green and yellow arrows indicate selective switch at 4 wavelengths; and d) structure of the MZ switch. (re-drawn from^[77]).

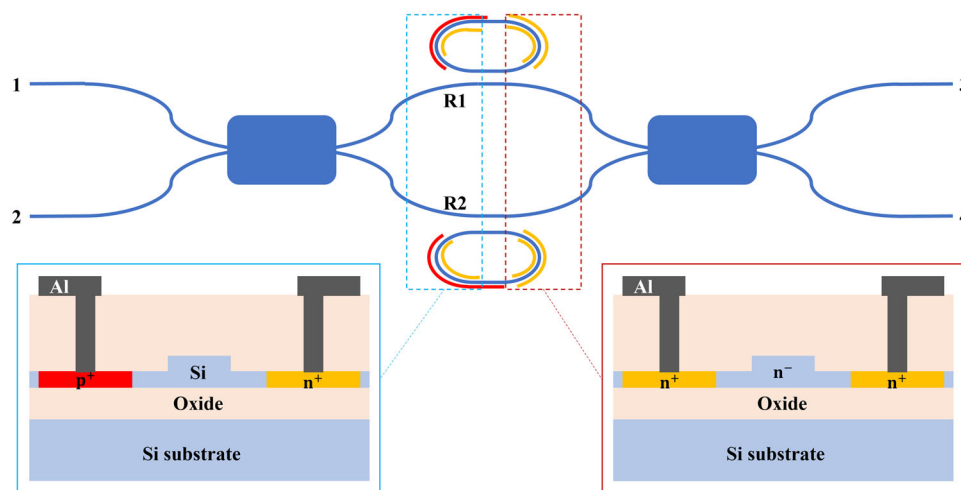


Figure 9. Structure of the 2×2 DR-MZI based E-O switch. Insets show the p-i-n diode and the silicon resistive microheater (cross-sectional views). (re-drawn from^[62]).

In 2014, Lu et al.^[62] proposed a 2×2 silicon integrated E-O switch based on double-ring assisted MZIs. The basic structure is shown in **Figure 9**, which is composed of a symmetric MZI and two racetrack microrings to combine the advantages of resonance enhancement in microrings and the coherent interference in MZIs. In the two micro-rings, both p-i-n diodes and silicon resistive microheaters are embedded for enabling active tuning. The p-i-n diodes are used for high-speed E-O switching and the microheaters are used for aligning the resonances of two micro-rings and tuning the switch operation wavelength. The double-ring assisted MZI (DR-MZI) is operated at the cross state when the resonance wavelength of two MRRs is aligned, since the phase difference between the two waveguide arms is zero. When the refractive index of one MRR is changed, the phase difference between the two arms changes to π and the switching state of the DR-MZI is at the bar state. This switch is fabricated using a CMOS compatible process on the SOI platform, which has 220 nm thick top silicon layer and 2 μm thick buried oxide layer. The radius of racetrack micro-ring is 10 μm , the coupling length between the racetrack micro-ring and the straight waveguide in the MZI is 2.9 μm , and the gap is 200 nm for achieving broadband operation. Results show that this device has a switching spectral window of 60 GHz together with a crosstalk better than -20 dB, and the E-O switch power from cross to bar states is 0.69 mW after phase error correction, which further consumes 2.3 mW T-O power. The rise time of switching is 405 ps and the fall time is 414 ps. The insertion loss ranges from 1.8 dB to 3.4 dB. Compared with the MRR only and MZI only structures, the DR-MZI structure achieves compact size and large operation bandwidth simultaneously. While a broader bandwidth of over 100 GHz has been shown using MRR, such as that achieved in ref. [75] reviewed in Section II.A, it is normally achieved by decreasing the MRR Q -factor. As a result, the passband roll-off is typically sacrificed, leading to a higher level of crosstalk. While the crosstalk performance can be improved by increasing the refractive index change, it may lead to higher insertion loss. On the other hand, the DR-MZI structure combines the advantage of both MRR and MZI structures, and hence, a broad operational

bandwidth can be achieved together with a low insertion loss and a low crosstalk.

Using a similar principle, a 4×4 silicon integrated optical switch based on DR-MZIs has been further demonstrated.^[63] The architecture is shown in **Figure 10**, which is a rearrangeable non-blocking switch with the 4×4 Benes switch fabric. There are a total of six switch elements organized in three stages, and each switch element is a 2×2 DR-MZI switch described above. Between these stages, 90° -crossed 1×1 MMIs, which are operated based on the self-imaging principle, are designed to cross the light over the waveguide junction while reducing the insertion loss and the waveguide crossing crosstalk. The 4×4 switch is fabricated, and it has a footprint of $3.4 \times 1.6 \text{ mm}^2$. Experimental results show that the device has an operational wavelength of 1555.62 nm, an on-chip insertion loss between 4.0 dB and 5.8 dB, a 35 GHz optical 3 dB bandwidth, and a crosstalk of lower than -18.4 dB. The minimum total T-O power consumption to align all resonances is 22.37 mW with a maximum E-O switching power consumption of 1.38 mW.

A larger scale 16×16 silicon integrated optical switch based on DR-MZIs was further reported by Guo et al.^[70] The 2×2 DR-MZI basic element is also further improved compared with previous studies, and the basic structure is shown in **Figure 11a**. It is still composed of a MZI with a microring coupled to each arm. As shown by the cross-sections of ring waveguides in the insets of the figure, a TiN microheater and a PIN diode are integrated on top of the upper micro-ring and across the upper ring waveguide, respectively while only a TiN microheater is integrated with the bottom micro-ring. Isolation trenches around the micro-rings are added for reducing thermal crosstalk. The T-O tuning of both micro-rings ensure that the resonances can be aligned or shifted to any arbitrary wavelength, and the T-O effect does not result in excessive loss. The fast switch between the two output ports is achieved by the E-O tuning of one micro-ring. With DR-MZI elements, the 16×16 optical switch is constructed using a Benes architecture, as shown in **Figure 11b**, which is composed of 56 DR-MZI switch elements. Each light path connecting the input port to the target output port passes through seven DR-MZI switch el-

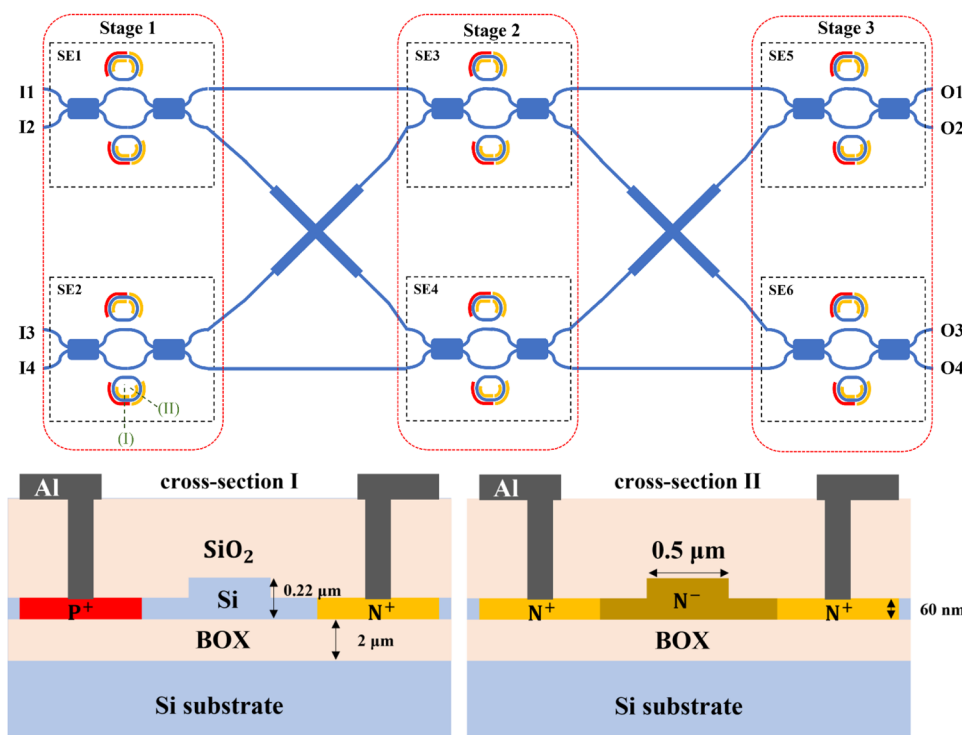


Figure 10. Architecture of the 4×4 Benes switch based on 2×2 DR-MZI elements. The structures of the p-i-n diode and the silicon resistive microheater are illustrated in cross-section views. (re-drawn from^[63]).

elements. Similar with the smaller scale demonstration,^[62] by controlling the resonances of two MRRs in a MZI, the cross state and the bar state can be switched. This switch is fabricated following similar parameters as the previous study,^[62] where the radius of the racetrack MRR is $10 \mu\text{m}$, and the coupling length and gap between the MRR and the MZI arm are $4.2 \mu\text{m}$ and $0.2 \mu\text{m}$, respectively. Measurement results show a crosstalk level of better than -20.5 dB at 1557.89 nm and an on-chip insertion loss of around $10.6 \pm 1.7 \text{ dB}$. The optical 3 dB bandwidth is around 0.33 nm , and the average E-O power consumption of the switch element is 0.34 mW . The 10%–90% rise time is measured to be 450 ps , and the corresponding fall time is 1.65 ns .

3. Phase Change Materials and Silicon Integrated Optical Switch with Phase Change Materials

While significant advances have been achieved in the past a few years, it is still challenging for silicon-integrated optical switches to achieve low power consumption, compact size, broad operation band, and low insertion loss simultaneously. Since PCM has characteristics that can be reversibly changed between two stable states at high speed with large differences in optical properties and these changes are self-holding, it has been considered as a promising option in the field of optical switches.^[98,99] In recent years, silicon-integrated optical switches further incorporating PCMs have been proposed and become a trending topic. VO_2 and GST are the most common PCMs used in silicon-integrated optical switches, and their phase change can be triggered by heating, electrical or optical ways.^[41,42,47,53–55] Therefore, in this section we review relevant recent studies, which are briefly summa-

rized in Table 2. Similar with the studies discussed in Section II, we also review silicon-integrated optical switches with PCMs according to the basic structure, where the MRR and the waveguide structures have been widely studied.

3.1. Silicon Integrated Optical Switches with PCM Based on the MRR Structure

In 2015, Sanchez et al.^[90] proposed a 2×2 silicon integrated optical switch incorporating VO_2 film. As shown in Figure 12, this switch is based on an add-drop ring resonator, where the waveguides are based on silicon while the micro-ring is based on VO_2 and silicon. The VO_2 film is deposited on top of the silicon ring with a spacer between them. When the VO_2 is in the insulating phase, the input signal is aligned with the resonance wavelength of the micro-ring, and hence, the optical signal is coupled via the ring and the switch is operated in the cross state (phase shift of 0). On the other hand, when the VO_2 film is in the metallic phase, the optical signal is not coupled to the micro-ring and the switch is operated in the bar state (phase shift of π). Therefore, by changing the real part of VO_2 refractive index, a π -phase shift can be achieved to switch the device between these two states. Based on this principle, results show that a footprint below $50 \mu\text{m}^2$, an insertion loss smaller than 3 dB and a crosstalk suppression ratio better than 10 dB in both states can be achieved.

To reduce the insertion loss of optical switch, Sanchez et al.^[91] further demonstrated another design of a 2×2 optical switch with VO_2 in 2016, which is shown in Figure 13. Similar with the previous work, the add-drop ring resonator structure is used, and

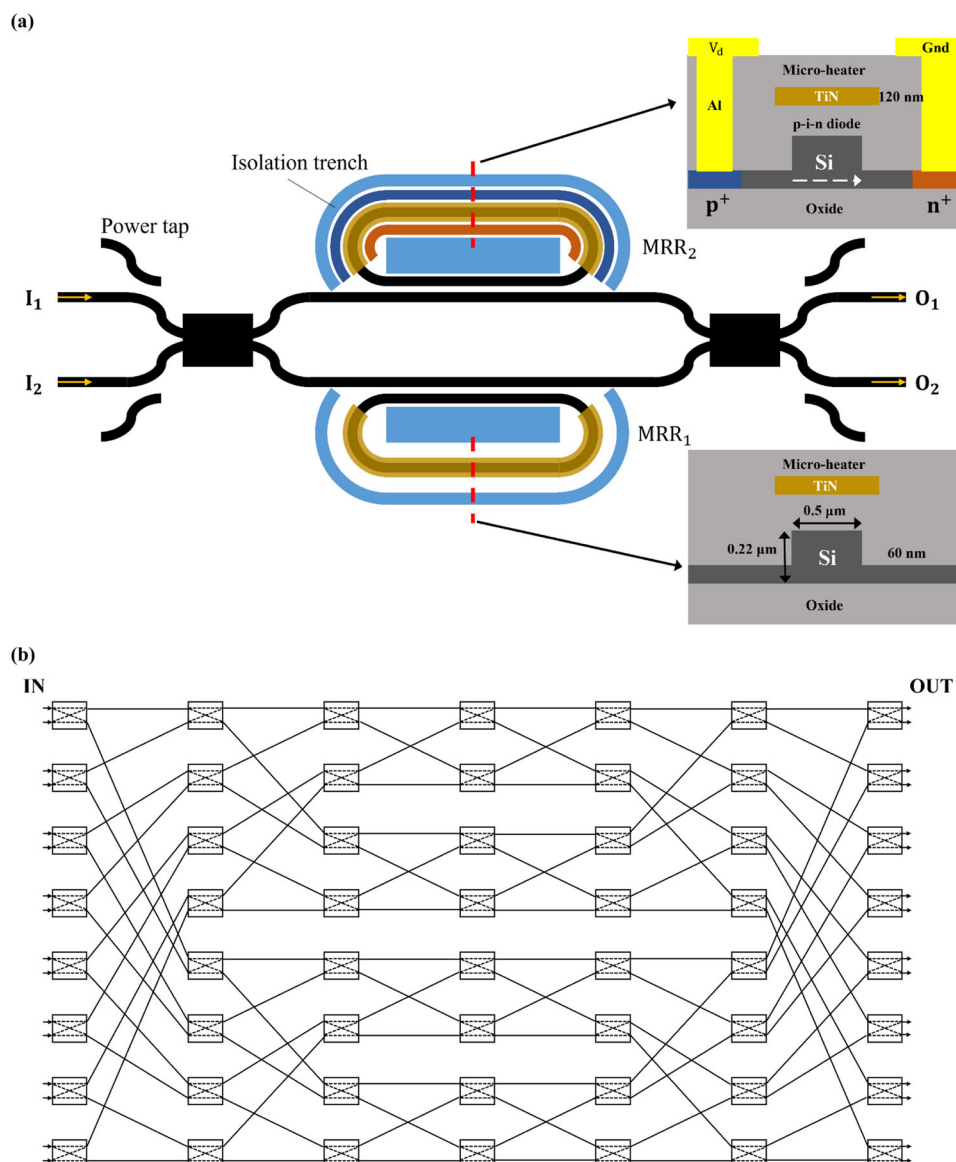


Figure 11. a) Structure of the 2×2 DR-MZI switch element. Insets show cross-sections of the upper micro-ring (top right) and the bottom micro-ring (bottom right); and b) topology of the 16×16 switch based on the Benes architecture. (re-drawn from^[70]).

the input and output waveguides are based on silicon while the micro-ring is based on VO_2 and silicon. This device also switches between the bar and cross states by exploiting the change of VO_2 complex refractive index. Different from the earlier work,^[90] here the VO_2 is only deposited over a part of the ring, which reduces the insertion loss, particularly when the VO_2 is under the high loss metallic phase. In this study, the hybrid VO_2/Si waveguide includes a standard single-mode silicon waveguide with a dimension of 480×220 nm, a silica cladding, and a VO_2 thin film layer, which has a thickness of 10 nm and is deposited on the top of the silicon waveguide and cladding. Metal contacts are then placed on both sides of the waveguide, as shown in Figure 13b, to change the VO_2 refractive index via electrooptical method. Results show that whilst the device footprint is slightly larger than the earlier work, this switch can achieve a crosstalk of above 12 dB

and an insertion loss below 1.8 dB, which are significantly improved compared with the prior work.^[90]

In addition to VO_2 , GST is another PCM that has been widely studied for optical switches in silicon-integrated circuits. Compared with VO_2 , GST has a higher phase transition temperature, so it is more suitable for high temperature environments. A wavelength-selective 2×2 optical switch based on a GST-assisted micro-ring was proposed by Zhang et al.^[94] in 2020, and its structure is shown in Figure 14. As shown in Figure 14a, this switch is composed of two crossed access optical waveguides, a micro-ring, and a bent GST-loaded silicon waveguide. The two access waveguides are crossed thereby allowing a relatively long coupling region to the silicon micro-ring. Further between the micro-ring and the bent GST-loaded silicon waveguide, there is a small uniform gap (W_g), forming a coupling region. Figure 14b shows the cross-

Table 2. Comparison of various silicon-integrated optical switches with phase change materials (MRR: micro-ring resonator; WG: waveguide; EXP: experimental; SIM: simulation; BW: bandwidth; ER: extinction ratio; IL: insertion loss; TE: transverse electric; and TM: transverse magnetic).

Year	Structure type	PCM	EXP/SIM	BW	Power	Speed	ER	Crosstalk	IL	Size
2015 ^[90]	MRR	VO ₂	SIM	–	–	–	–	>10 dB	<3 dB	<50 μm ²
2016 ^[91]	MRR		SIM	–	–	–	–	>12 dB	<1.8 dB	<100 μm ²
2018 ^[92]	WG		EXP	>30 nm	TE: 45 mW TM:70 mW	–	>20 dB	–	<1 dB	–
2021 ^[93]	WG		EXP	–	–	318 ns	0.7 dB μm ^{−1}	–	–	–
2020 ^[94]	MRR	GST	SIM	–	–	–	About 20 dB	–	0.9 dB at the through port, 2 dB at the drop port	–
2020 ^[95]	WG		EXP	2.2 nm	9.5 nJ and 3.2 nJ	–	>15 dB	–	<5 dB	–
2021 ^[96]	MRR		SIM	–	–	–	OFF state: 19 dB, ON state: 10 dB	–	OFF state: 1.52 dB, ON state: 2.02 dB	–
2021 ^[97]	WG		EXP	–	3.528 pJ	1.5 ns	About 44 to 46 dB	–	–	–

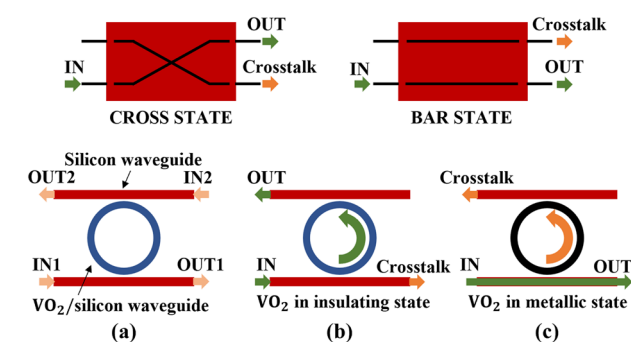


Figure 12. Structure of silicon integrated optical switch incorporating VO₂. a) Schematic of the switch based on an add-drop ring resonator; b) the “cross” switching state; and c) the “bar” switching state. (re-drawn from^[90]).

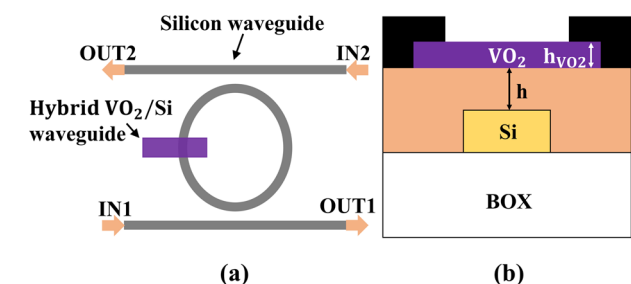


Figure 13. a) Structure of the 2 × 2 switch based on an add-drop MRR (partially covered with VO₂); and b) the cross-section view of the hybrid waveguide. The electrodes are represented in black. (re-drawn from^[91]).

section of the GST-loaded silicon waveguide, which is formed by depositing a GST thin layer on top of a silicon strip waveguide. The silicon microring and the GST-loaded bent silicon waveguide in the coupling region are concentric, and their key parameters, such as widths and bending radii, can be selected following the optimal phase-matching condition. When the GST is in the crystalline state (ON state), the input signal coupled to the micro-ring

is further strongly coupled to the bent GST-loaded silicon waveguide. Due to the high-loss in GST-loaded waveguide, the resonance peak at the drop port of the switch is suppressed significantly. Correspondingly, there is a low loss transmission for the resonance wavelength at the through port. On the other hand, when the GST is changed to the amorphous state (OFF state), there is significant phase mismatch between the micro-ring and the GST-loaded silicon waveguide, and hence, there is almost no coupling between them. Therefore, at the resonance wavelength, the signal can propagate to the drop port with a low transmission loss. Thus, a low-loss wavelength selective optical switch can be realized by modifying the GST state. According to the 3D finite-difference method simulation result, this switch has about 20 dB extinction ratio at the resonance wavelength. The excess loss is 0.9 dB at the through port and 2 dB at the drop port. Compared with previous studies reviewed above,^[90,91] both the extinction ratio and insertion loss performances have been improved.

In 2021, a 1 × 2 silicon integrated optical switch with GST was further proposed by Mahmoodi et al.^[96] The structure is shown in Figure 15. In this design, instead of depositing PCM on top of the silicon waveguide^[90,91] or adding an additional PCM-loaded waveguide,^[94] a sector of the silicon microring is replaced with a GST-filled slot waveguide. When GST is in the amorphous phase (OFF state of this switch), the input light with a wavelength equal to the resonant wavelength of the micro-ring is coupled from the top input waveguide via the micro-ring to the bottom waveguide and further transferred to the cross output port (T2 port in Figure 15). The real and imaginary parts of GST refractive index are changed when an electric field with sufficiently large amplitude is applied. In this case, the resonance wavelength of the microring is changed, and an insignificant amount of input power is coupled to the micro-ring. In this case (ON state), the input signal mainly propagates directly to the through port (T1 port in Figure 15). In the OFF state, the extinction ratio of this optical switch is shown to be 19 dB and the insertion loss is 1.52 dB. In the ON state, the extinction ratio and the insertion loss of this optical switch is 10 dB and 2.02 dB, respectively. Due to the use of GST-filled slot waveguide, the signal light directly interacts with

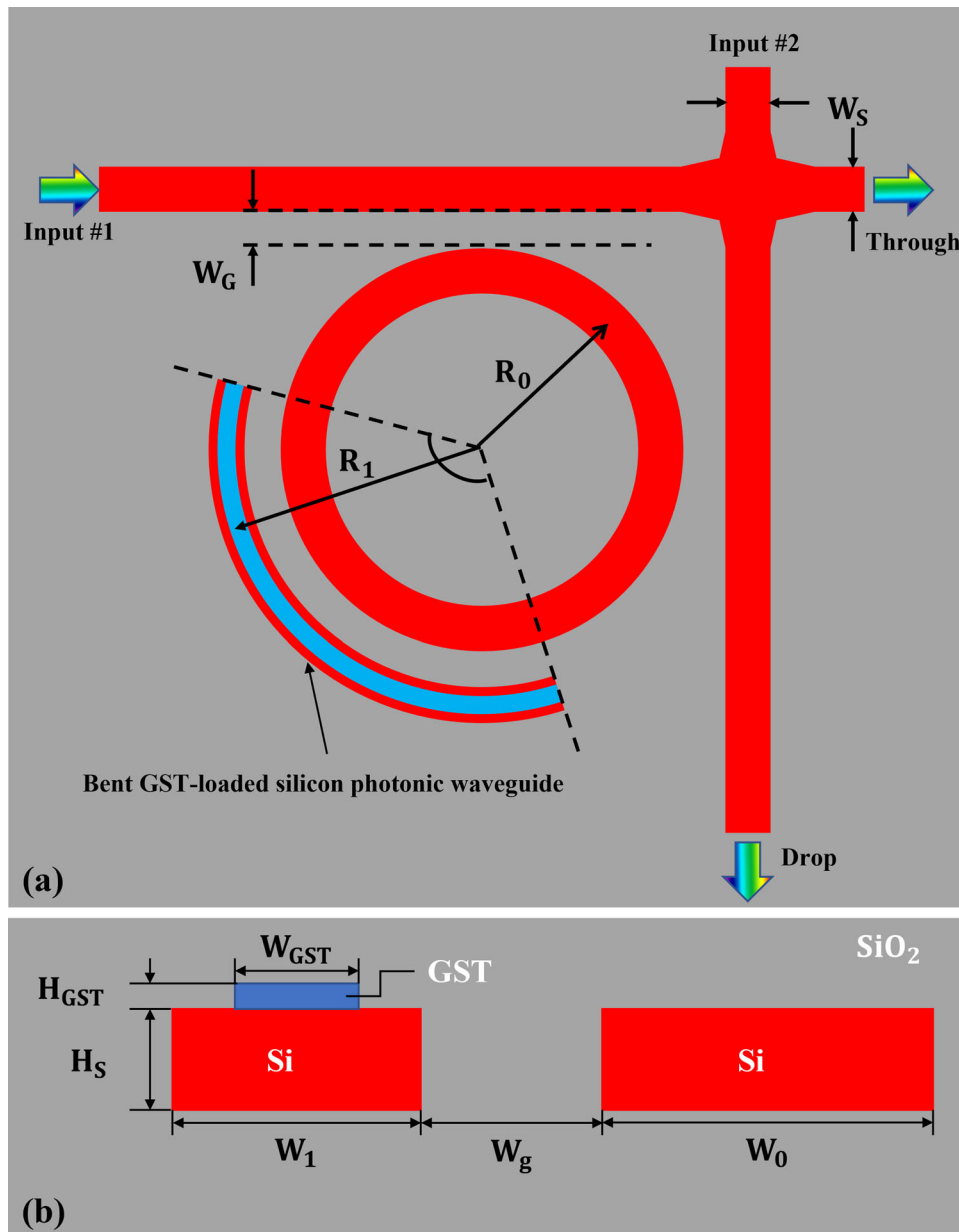


Figure 14. Structure of the wavelength-selective photonic switch based on a GST-assisted microring. a) Top view of the device; and b) cross section view of the coupling region. (re-drawn from^[94]).

the PCM. Hence, the state change of GST has a more direct and larger impact on the signal optical field, thereby realizing a highly efficient optical switch.

3.2. Silicon Integrated Optical Switches with PCM Based on the Waveguide Structure

In addition to the PCM-based silicon-integrated optical switch adopting the MRR structure, the waveguide structure has also been widely explored. In the waveguide-based switches, instead of switching the input signal to different output ports, normally the on-off switching function is explored. In 2018, Oli-

vares et al.^[92] demonstrated a silicon-integrated optical switch with VO_2 that was thermally triggered by lateral microheaters. The concept of the hybrid VO_2/Si waveguide, which has a lateral Ti microheater based on a double metallization process integrated, is shown in **Figure 16**, and the cross section of the hybrid VO_2/Si waveguide is shown in the inset. The hybrid VO_2/Si waveguide is inserted into a straight silicon waveguide, and VO_2 is then deposited on top of the silicon waveguide, which has a dimension of $480 \text{ nm} \times 220 \text{ nm}$. A spacer is added between the silicon waveguide and the VO_2 layer, which is composed of a 10 nm thick oxide plus a 50 nm thick nitride (SiN) hard mask. The SiN layer is used to planarize and protect the silicon surface. A microheater is further added on top of the VO_2 layer laterally, and

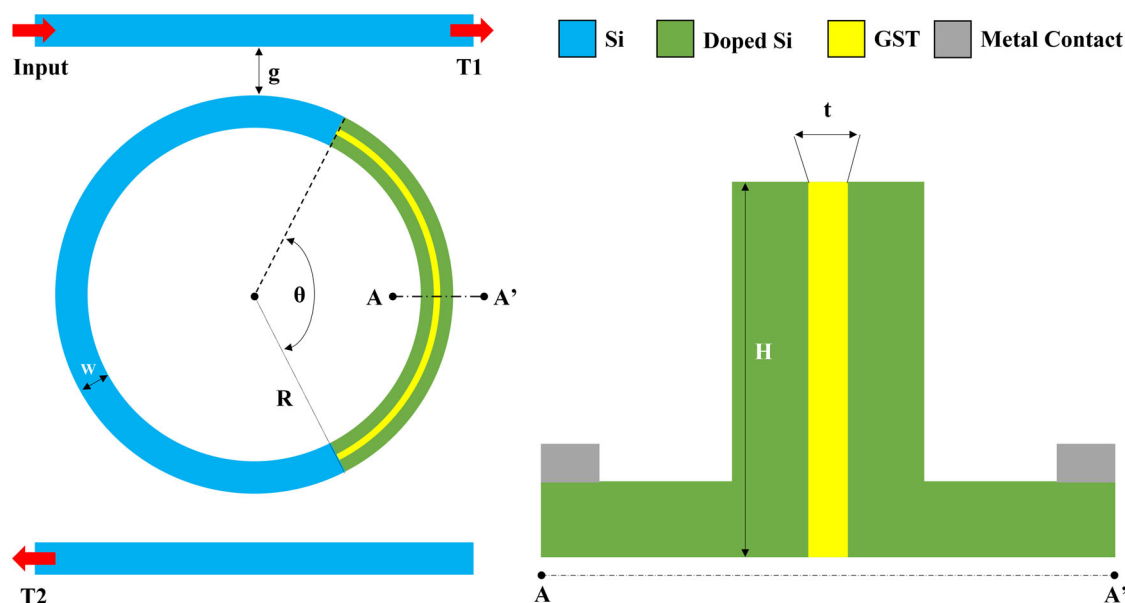


Figure 15. Structure of the 1×2 switch with GST-filled slot waveguide. (re-drawn from^[96]).

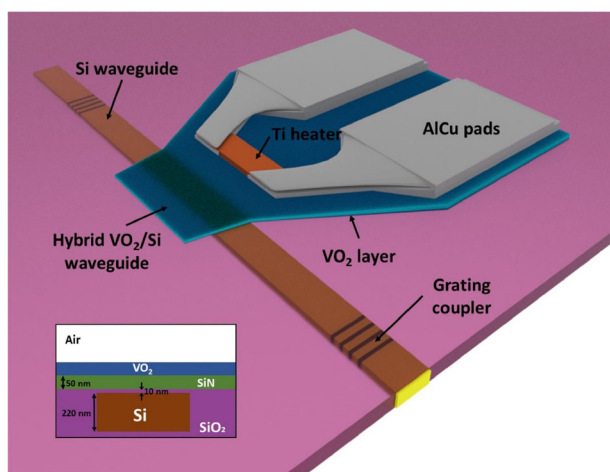


Figure 16. Structure of the hybrid VO_2/Si waveguide. A lateral Ti micro-heater is integrated on top. The inset shows the cross section of the hybrid VO_2/Si waveguide. Reproduced under terms of the CC-BY license.^[92] Copyright 2022, Optica Publishing Group.

the generated heat by the microheater can be efficiently delivered to the hybrid waveguide. Therefore, the VO_2 layer on top of the silicon waveguide can be switched between the insulating and the metallic states in a very short length to change the optical loss significantly to realize on-off switching. This switch is fabricated on the SOI platform with a $2 \mu\text{m}$ buried oxide layer. The silicon waveguide structures are first fabricated by the standard process, and then the wafer is polished by chemical mechanical planarization (CMP). After that the wafer is diced, and molecular beam epitaxy (MBE) is used to grow a 40 nm thick amorphous VO_x layer. The VO_x deposited on the undesired regions is then removed by the lift-off process. Finally, to form polycrystalline VO_2 layer, the ex-situ annealing process is used. In addition, before the

metal evaporation and lift-off processes, the electrodes are fabricated by two e-beam positive resist exposures, where a 90 nm thick Ti is processed first, followed by depositing 400 nm thick AlCu pads. Experimental results show that this optical switch has an insertion loss lower than 1 dB and an extinction ratio higher than 20 dB in a broad wavelength region exceeding 30 nm. In addition to the low loss, high extinction ratio and broad bandwidth, compared with previous MRR-based structures, this waveguide-based switch also supports both TE and TM polarizations, where the electrical switch power consumption is around 45 mW and 70 mW, respectively.

The studies reviewed above rely on either the thermal tuning or the electrical activation of PCMs. Parra et al.^[100] reported an integrated all-optical waveguide switch using VO_2 in 2020. The cross-section of the hybrid VO_2/Si waveguide is shown in **Figure 17**. The silicon waveguide size is $480 \text{ nm} \times 220 \text{ nm}$, thereby ensuring the single-mode operation. To planarize and protect the silicon surface, a 50 nm thick SiN layer is used, with a 10 nm thick oxide between the silicon waveguide and the SiN layer. At the top of the SiN layer, a 40 nm thick VO_2 layer is deposited. Different from previous works, here a square optical pump pulse is used to trigger the phase transition and to realize the all-optical on-off switch. Experimental investigation shows that it has 350 ns and 7 μs switching speed for the insulating-to-metallic transition and the metallic-to-insulating transition, respectively. Following this work, Parra et al.^[93] in 2021 conducted further device testing and characterization. Experimental results show that based on a maximum switchable length of $15 \mu\text{m}$, this all-optical waveguide switch achieves an extinction ratio of $0.7 \text{ dB } \mu\text{m}^{-1}$, an optimal switching speed of 318 ns, and an energy consumption per switch of 15.8 nJ.

It is worth mentioning that while nonvolatile phase change has been achieved in VO_2 , the integrated optical switch studies using VO_2 reviewed above do not feature the latching property. For example, a continuous power consumption is needed to main-

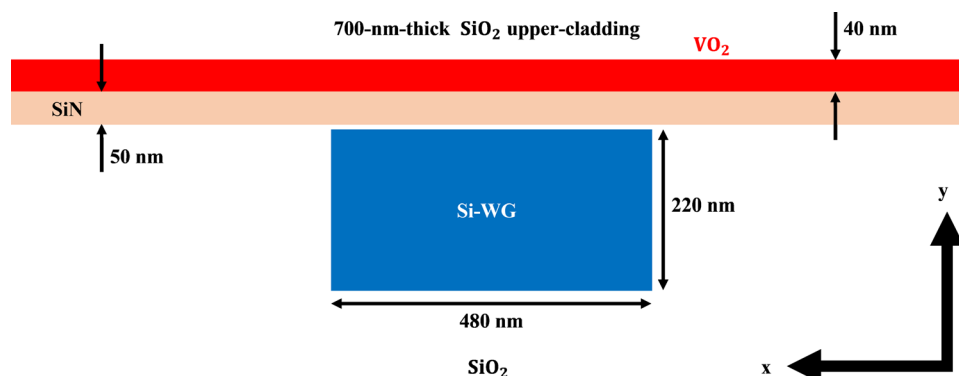


Figure 17. Cross section view of the all-optical switch based on hybrid VO_2/Si waveguide. (re-drawn from^[100]).

tain the operation temperature requirement in the thermal-tuned VO_2 switches.^[100] Although tuning the oxygen vacancy concentration of VO_2 can achieve non-volatile and reversible phase transitions, the process is too slow for practical switching operations. Recently, Jung et al. proposed and demonstrated an optical memory based on the VO_2 microwire integrated on a Si photonic waveguide.^[101] This device uses the bistable characteristic of VO_2 and realizes the change of optical transmission power of the Si waveguide, achieving effective switching operation. While a very low power consumption of 23.5 pJ per switch has been demonstrated, it is still volatile as a voltage bias is needed to maintain the IMT hysteresis. However, this work provides a promising solution to non-volatile VO_2 -based integrated optical switching by further exploring the narrow hysteresis of VO_2 reported by the same group in 2021.^[102] While more detailed studies are required to develop and demonstrate a fully functional silicon-integrated device, the non-volatile property of VO_2 at room temperature using the first major transition between insulator state and oscillating state reported for the first time paves the foundation to unleash the latching property of VO_2 -based integrated optical switches.^[102]

An electrical-driven phase change optical switch consisting of a Si waveguide and GST film with ITO or graphene heaters was reported by Li et al.^[97] in 2021. Different from the previous works,^[92,93,100] which use VO_2 as the PCM and use Ti heater or square optical pump pulse to trigger the phase transition, this optical switch uses GST as PCM and uses ITO or graphene heaters to trigger the phase transition. The structure and the switching principle of the switch are shown in **Figure 18**. **Figure 18a** illustrates that the proposed switch consists of a GST-on-silicon hybrid waveguide. The GST thin film, which has a width of 500 nm, a thickness of 30 nm, and a length of 2 μm , is added on top of a rectangular Si waveguide with the same width of 500 nm. The electrodes, which have a thickness of 20 nm, are distributed on both sides of the part of waveguide that is covered by GST for conducting the electrical switching signal, where one is applied with the electric pulse and the other is grounded. A heat transfer layer is added underneath the electrodes to improve the electrical contact. Different electrode materials are compared, including Au, Cu and Pd, and two heater materials are also studied, namely ITO and graphene. It is shown that the electrode material has little effect on the phase change of GST, so low-cost electrodes, such as copper, can be used. On the other hand, the heater mate-

rial largely affects the GST phase change, where it is found that the switch with graphene has better performance than ITO.

Figure 18b further shows the operation principle of the switch. For the GST crystallization process, a single pulse with low voltage and long pulse width is applied, which can heat GST above the crystallization threshold temperature ($T_g = 433$ K) but below the melting temperature ($T_m = 893$ K), for ensuring the complete crystallization of GST and achieving a large refractive index change. On the other hand, for the GST amorphization process, a single pulse with high voltage and short pulse width is applied to the electrode, for heating GST above the melting temperature. This pulse is then immediately removed to achieve the quenching effect, which enables GST to return to the amorphous state with low optical constant. This process ensures the reversible conversion of the switch. Frequency domain finite element method (FEM, based on COMSOL Multiphysics) is used to simulate the device performance. Results show that in the case of amorphous GST (a-GST), because of the low refractive index, the optical signal is almost fully bounded in the Si waveguide. Therefore, the input light propagates through the Si-GST hybrid waveguide with relatively small optical attenuation, leading to the “ON” state of the switch. In the case of crystalline GST (c-GST), because of the large refractive index, the electric field of the light wave penetrates to the GST film significantly. Hence, the light propagation is strongly attenuated and almost no optical signal propagates to the output port, resulting in the “OFF” state. In terms of the optical performance of the switch, an extinction ratio of 44 to 46 dB in a spectral range of 1525 to 1575 nm is achieved. Compared with prior works, the extinction ratio achieved is much larger. In addition, its energy consumption is 3.528 pJ and its switching speed is 1.5 ns.

In addition to the straight waveguide structure, Hu et al.^[95] presented a 1×2 wavelength-selective optical switch based on the Si-GST grating assisted waveguide structure in 2020. Different from the other waveguide-based works using PCM,^[92,93,97,100] this optical switch further introduces the grating structure, thereby enabling the function of path selection instead of simple on-off switch. The structure is shown in **Figure 19**, and it is consisted of two parallel silicon strip waveguides. One waveguide has periodic inner sidewall corrugations with 50% duty cycle. On top of the corrugated waveguide, GST nano-strips is integrated. The straight silicon waveguide has a width (W_1) of 370 nm, and the grating-assisted waveguide (loaded with GST) has a basic width

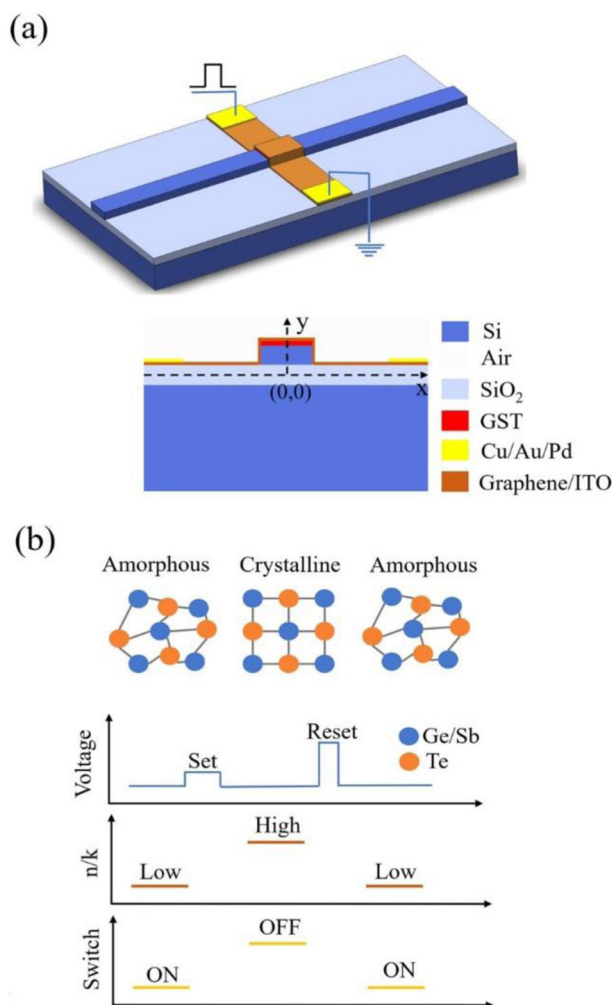


Figure 18. The structure and switching principle of integrated PCM switch with ITO/graphene heaters. a) The 3D structure and the cross-section view of the switch; and b) the operation principle of the switch. Reproduced with permission.^[97] Copyright 2022, IOP Publishing Ltd.

(W_2) of 400 nm. The GST grating period (Λ) is 374 nm, corresponding to a central wavelength of 1553 nm. The GST nanostrip has a width W_{GST} of 390 nm and a length L_{GST} of 100 nm. Figure 19b illustrates the cross-sectional view of the device, where the silicon waveguide height (H) is 220 nm, the corrugation strength (D) is 50 nm, the gap between the two waveguides (G) is 230 nm, and the GST layer thickness (H_{GST}) is 50 nm. When GST is in the amorphous state, due to the phase matching condition being satisfied, the input light is reflected to the drop port. When the GST changes from the amorphous state to the crystalline state, there is a large change in the GST refractive index, leading to the redshift of the Bragg wavelength and the break of phase matching condition. In the meanwhile, the attenuation coefficient of GST also increases substantially, resulting in a higher loss of the GST-loaded silicon waveguide. Therefore, the input light passes to the through port directly while the coupling to the drop port is suppressed. The device is fabricated based on the SOI platform, where a 50 nm thick GST film is deposited using radio frequency (RF) sputtering from a stoichiometric $\text{Ge}_2\text{Sb}_2\text{Te}_5$ al-

loy target, and the lift-off process in a warm acetone bath is used to pattern the GST layer. Experimental results show that the Si-GST grating assisted switch has a central operation wavelength of 1576 nm, a <5 dB insertion loss, a >15 dB extinction ratio, and an operation bandwidth of 2.2 nm. The reversible phase change of GST can be obtained by controlling the pulse energy, which is 9 aJ nm^{-3} for the GST phase transition from crystalline to amorphous, and 3 aJ nm^{-3} for the GST phase transition from amorphous to crystalline. Therefore, the energy consumption for this device with 540 GST cells is around 9.5 and 3.2 nJ, respectively.

4. Conclusions and Discussions

With the end of Moore's law and the slowdown of electronic technology, there is an increasing need to find alternative solutions. Integrated silicon photonic technology has the potential to attain faster operation speed, lower power consumption, and higher data density, and it also has CMOS compatibility.^[103–105] Hence, it has become a highly promising alternative. As a key device in silicon-integrated photonic circuits, the integrated optical switch has a wide range of applications and has been extensively studied.^[8–14,16–18]

As reviewed in Section II, currently the MZI and the MRR are the most common structures for silicon-integrated optical switches. However, some of their shortcomings limit further advancements. For the MZI structure, which is a mature structure,^[106] the large footprint is the main disadvantage. The power consumption is also relatively high, making it unfavorable for large-scale integrations.^[82,107–109] In addition, whilst it can support relatively large operation bandwidth, it has limited wavelength selectivity. On the other hand, the MRR structure has advantages of compact footprint, wavelength selectivity, and relatively low power consumption.^[60,110–112] However, it also has the disadvantages of narrow bandwidth, temperature susceptibility, and requiring high processing accuracy.^[113–115] This is in contrast with the typical demands for broad operation bandwidth and stable operation, such as allowing the use of multiple signal wavelengths to increase the overall capacity of data transfer, which is important to support the increasing volume of data that needs to be processed and interconnected. In addition, while the hybrid type of silicon integrated optical switches combining MRR and MZI has been further proposed and investigated, they still have relatively narrow bandwidth^[116,117] and large insertion loss.

Although significant advancements have been achieved in silicon-integrated optical switches, they normally require high power consumption and have the thermal crosstalk issue when the T-O effect is used. On the other hand, when the T-O effect is explored, integrated optical switches rely on a relatively weak change of refractive index that introduces further losses. In addition, silicon-integrated optical switches are non-latching (nonvolatile) and require continuous power consumption just to maintain the switch status.^[36] To overcome these limitations, PCMs have been introduced on silicon to realize integrated optical switches. As discussed in detail in Section III, the waveguide and the MRR are the most common structures when PCMs are used. However, the waveguide option typically requires long length, has relatively large loss, and does not have the wavelength selectivity to provide flexible routing that is highly demanded in interconnect applications, particularly when the wavelength-

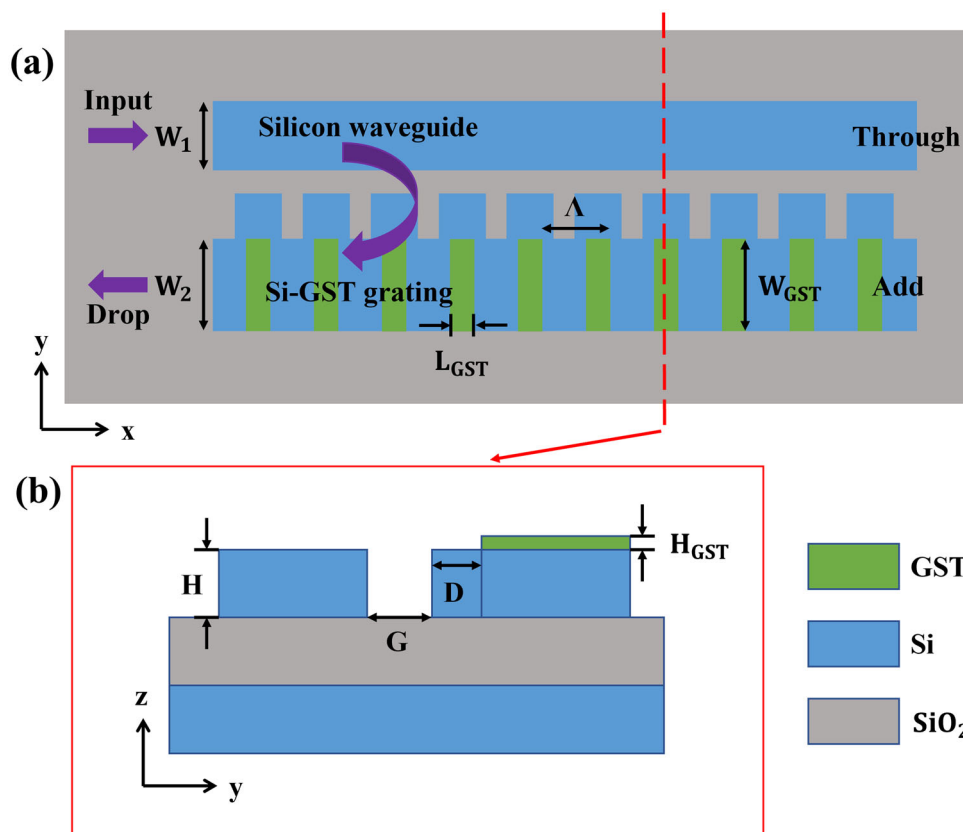


Figure 19. Structure of the Si-GST grating assisted waveguide switch. a) The top view of the device; and b) the cross-sectional view along the dashed line in (a). (re-drawn from^[95]).

division-multiplexing (WDM) is used to increase the overall data rate. In addition, the waveguide type is mainly used for the on-off type of switch, and only limited study has further realized the switch between different signal paths. On the other hand, the MRR structure with PCMs have similar disadvantages as those based on pure silicon, such as the narrow bandwidth.

While silicon-integrated optical switches with PCMs have seen significant progress during the past a few years, so far, most studies are only based on simulations and such devices have not been fabricated and tested. Therefore, in future studies, the development of reliably, repeatable, and high-quality fabrication processes for PCMs need to be further studied and the experimental verification of the proposed optical switches is also needed. In addition, almost all integrated optical switches with PCMs that have been investigated are 1×1 , 1×2 , or 2×2 , and large-port-count optical switches incorporating PCMs have not yet been seen. However, as the architecture of silicon-integrated photonic circuits becomes increasingly complicated and with the popularity of distributed signal processing and computation in integrated circuits (e.g., multi-core processing), the optical signal needs to be switched to a large number of destinations with high efficiency. Therefore, there are high demands for the PCM-based integrated switches to support large-port-count and non-blocking signal switching, which also require further study.

In terms of the types of PCMs that have been explored in silicon-integrated optical switches, VO_2 and GST are the com-

mon materials. GST's crystallization temperature (i.e., the phase transition temperature) is about 160°C (433 K) and the amorphization temperature (i.e., the melting temperature) is about 600°C (873 K). Hence, a relatively high power consumption is needed to drive its phase transition. On the other hand, VO_2 has a much lower phase transition temperature of around 340 K. However, it is too close to the room temperature, requiring highly efficient heat dissipation that limits its application in high-temperature environments. Furthermore, the insulator-metal transition of VO_2 is volatile. Although tuning the oxygen vacancy concentration using a gated electrolyte or hydrogenation of VO_2 can enable VO_2 to achieve non-volatile and reversible phase transitions, this process is very slow.^[102] Considering these fundamental limitations, other types of PCMs also worth further investigations, such as $\text{Ge}_2\text{Sb}_2\text{Se}_4\text{Te}_1$ (GSST), Sb_2S_3 , and Sb_2Se_3 .^[101] The GSST combines broadband transparency ($1\text{--}8.5\ \mu\text{m}$) and large optical contrast ($\Delta n = 2.0$). Compared with GST, GSST shows three orders of magnitude lower absorption coefficient while preserving a large Δn of 2.1 to 1.7 across the near- to mid-IR bands.^[118] For the Sb_2S_3 , the imaginary part of the refractive index is very small at selected wavelengths that are widely used in optical communications, such as 1060 nm, 1310 nm and 1550 nm. In particular, at the most widely used wavelength of 1550 nm, its refractive index is $2.712 + 0i$ in amorphous phase and $3.308 + 0i$ in crystalline phase. Sb_2Se_3 has similar property. At 1550 nm wavelength, its refractive index is

3.285 + 0i in the amorphous phase and 4.050 + 0i in crystalline phase.^[119] Hence, compared with GST, due to the zero imaginary refractive index, Sb₂S₃ and Sb₂Se₃ show lower absorption in both phases, providing the possibility to realize low loss integrated optical switches. Comparing these two materials, Sb₂S₃ has lower losses in general, but it has issues where the crystallization is not homogeneous, and the damage threshold is close to the crystallization temperature. On the other hand, the Sb₂Se₃ has a larger real refractive index Δn (i.e., stronger light confinement), and a higher durability of phase switching,^[120] providing promising features. Whilst these new types of PCMs have attractive characteristics, detailed studies are needed to investigate their capability for integrated optical switches.

Furthermore, whilst PCMs has the advantage of nonvolatile, which indicates that it is possible to maintain the state of the material and the corresponding switching state without a continuous supply of energy, this unique property is still to be studied and demonstrated in more detail in silicon integrated optical switches. This highly desired property to realize low power consumption has been discussed in more detail in ref. [121] which provides a structured and comprehensive review on the possible ways to achieve non-volatile switching in silicon photonic devices, such as the optical bistability induced by thermal-optic effect, the resistive switching, and the use of MEMS structures.

This paper focuses on the review of basic silicon-integrated optical switch devices. The application of these silicon-integrated optical switches has also attracted intensive attention recently, mainly focusing on optical communication and information processing. For example, Nakamura et al. demonstrated a silicon-integrated reconfigurable optical add-drop multiplexer (ROADM) based on an 8 × 8 integrated optical switch.^[122] Such integrated ROADM with compact size is highly demanded in optical communication systems and networks with WDM, to achieve flexible wavelength allocation and more efficient use of wavelength resources in both core networks and access networks. In addition, silicon-integrated optical switches have also been used as core devices in programmable photonic integrated circuits, which can enable diverse signal processing functions to be realized within a single chip, such as the implementation of machine learning signal processing algorithms in the optical domain.^[123] With the recent progress of silicon-integrated optical switches, particularly the possibility of achieving low power consumption latching switches, more applications can be enabled in the future.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

Mach-Zehnder interferometers, microring resonators, optical switches, phase-change materials

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